

ASSIGNMENT-7.1

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BATCH-46

Task Description #1 (Syntax Errors — Missing Parentheses in Print Statement)

Task: Provide a Python snippet with a missing parenthesis in a print statement (e.g., print "Hello"). Use AI to detect and fix the syntax error.

Bug: Missing parentheses in print statement def greet():

```
    print "Hello, AI Debugging Lab!"
```

```
    greet()
```

Requirements:

- Run the given code to observe the error.
- Apply AI suggestions to correct the syntax.
- Use at least 3 assert test cases to confirm the corrected code works.

Expected Output #1 :

- Corrected code with proper syntax and AI explanation.

PROMPT:

#Detect and fix the syntax error in this Python code:

#def greet(): print 'Hello, AI Debugging Lab!' ; greet() #Explain the issue, provide corrected Python 3 code, and include 3 assert tests to verify correct output.

CODE:

```
def test_greet():
```

```
    captured_output = io.StringIO()      #      Create      StringIO
```

```
    object sys.stdout = captured_output # Redirect stdout.
```

```
    greet()      # Call the function.  sys.stdout =
```

```
    sys._stdout_ # Reset redirect.
```

```
assert captured_output.getvalue().strip() == 'Hello, AI Debugging Lab!'
test_greet()
```

```
def test_greet_output():
```

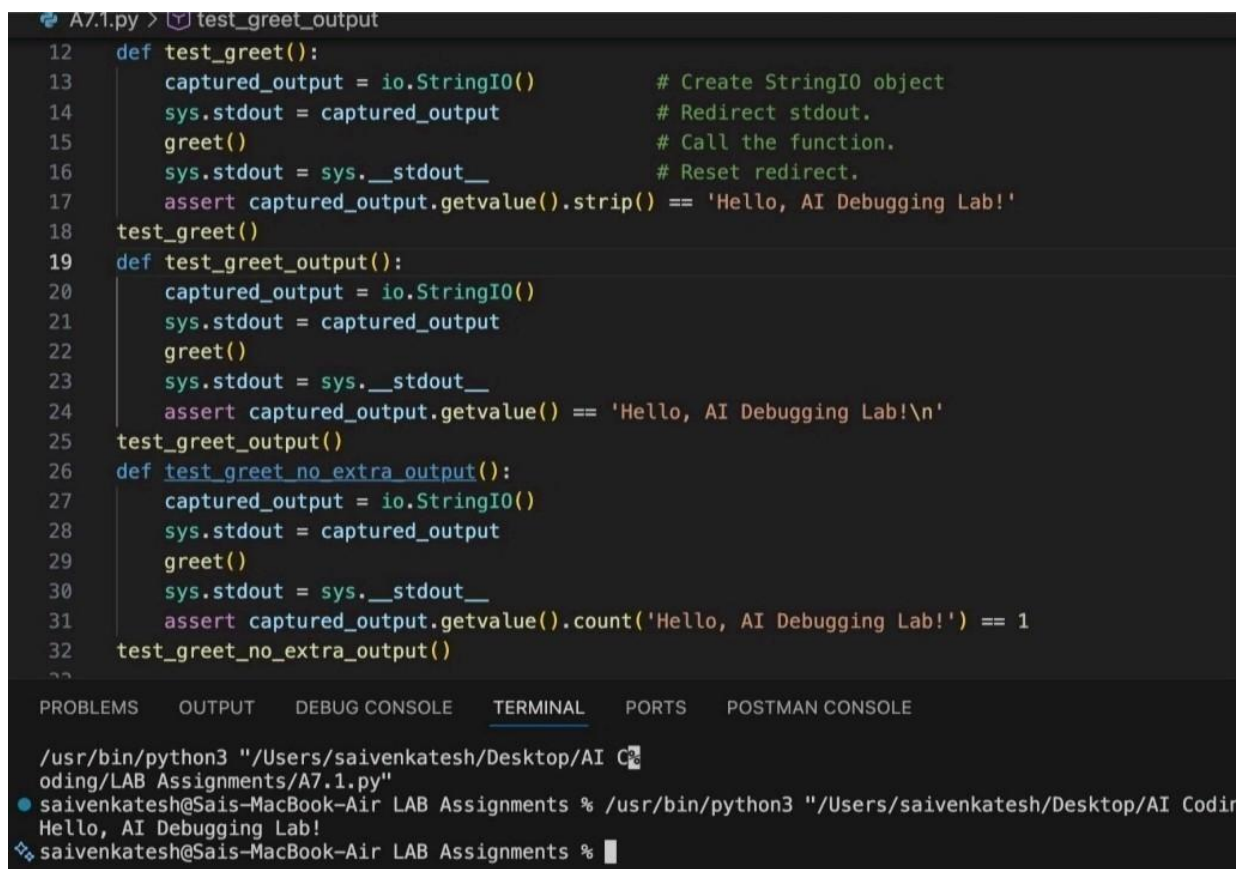
```
    captured_output = io.StringIO() sys.stdout = captured_output greet()
    sys.stdout = sys.__stdout__ assert
    captured_output.getvalue() == 'Hello, AI Debugging Lab!\n'
```

```
test_greet_output() def test_greet_no_extra_output():
```

```
    captured_output = io.StringIO() sys.stdout = captured_output greet() sys.stdout
    = sys.__stdout__ assert
```

```
captured_output.getvalue().count('Hello, AI Debugging Lab') == 1
```

```
test_greet_no_extra_output() output:
```



```
A7.1.py > test_greet_output
12 def test_greet():
13     captured_output = io.StringIO()           # Create StringIO object
14     sys.stdout = captured_output              # Redirect stdout.
15     greet()                                   # Call the function.
16     sys.stdout = sys.__stdout__              # Reset redirect.
17     assert captured_output.getvalue().strip() == 'Hello, AI Debugging Lab!'
18 test_greet()
19 def test_greet_output():
20     captured_output = io.StringIO()
21     sys.stdout = captured_output
22     greet()
23     sys.stdout = sys.__stdout__
24     assert captured_output.getvalue() == 'Hello, AI Debugging Lab!\n'
25 test_greet_output()
26 def test_greet_no_extra_output():
27     captured_output = io.StringIO()
28     sys.stdout = captured_output
29     greet()
30     sys.stdout = sys.__stdout__
31     assert captured_output.getvalue().count('Hello, AI Debugging Lab!') == 1
32 test_greet_no_extra_output()

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

/usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/LAB Assignments/A7.1.py"
saivenkatesh@Sais-MacBook-Air LAB Assignments % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/LAB Assignments/A7.1.py"
Hello, AI Debugging Lab!
saivenkatesh@Sais-MacBook-Air LAB Assignments %
```

Explanation:

The issue in the original code is that it uses Python 2 syntax for the print statement. In Python 3, print is a function and requires parentheses.

Task Description #2 (Incorrect condition in an If Statement)

Task: Supply a function where an if-condition mistakenly uses = instead of Let AI identify and fix the issue.

```
# Bug: Using assignment (=) instead of comparison (==)
def check_number(n):
    if n = 10:
        return "Ten"
    else:
        return "Not Ten"
```

Requirements:

- Ask AI to explain why this causes a bug.
- Correct the code and verify with 3 assert test cases.

Expected Output #2:

- Corrected code using == with explanation and successful test execution.

Prompt used:

Detect and fix the error in this Python code: def check_number(n): if n = 10: return "Ten" else: return "Not Ten". #Explain why using causes a bug, provide corrected code using '==', and include 3 assert tests to verify the function works. CODE:

```
def check_number(n):
    if n == 10:
        return "Ten"
    else:
        return "Not Ten"
```

Assert tests

```
assert check_number(10) == "Ten"
assert check_number(5) == "Not Ten"
assert check_number(0) == "Not Ten"
print("All tests passed! ")
```

```
A7.1.py > ...
39 #Detect and fix the error in this Python code: def check_number(n): if n
40 #Explain why using '=' causes a bug, provide corrected code using '==',
41 # The error occurs because '=' is an assignment operator, not a comparis
42 # In Python, '==' must be used inside conditional statements.
43
44 def check_number(n):
45     if n == 10:
46         return "Ten"
47     else:
48         return "Not Ten"
49
50 # Assert tests
51 assert check_number(10) == "Ten"
52 assert check_number(5) == "Not Ten"
53 assert check_number(0) == "Not Ten"
54
55 print("All tests passed!")
56
57
58
59
```

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```
/usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/LAB Assignments/A7.1.py"
saivenkatesh@Sais-MacBook-Air LAB Assignments % /usr/bin/python3 "/Users/saivenk
All tests passed!
saivenkatesh@Sais-MacBook-Air LAB Assignments %
```

Explanation :

The error occurs because = is an assignment operator, not a comparison operator. In an if condition, Python requires == to compare values, and using = causes a SyntaxError. Replacing = with == correctly checks whether n is equal to 10, allowing the function to work properly.

Task Description #3 (Runtime Error — File Not Found)

Task: Provide code that attempts to open a non-existent file and crashes. Use AI to apply safe error handling.

Bug: Program crashes if file is missing def read_file(filename): with open(filename, 'r') as f: return f.read() print(read_file("nonexistent.txt")) Requirements:

- Implement a try-except block suggested by AI.
- Add a user-friendly error message.

- Test with at least 3 scenarios: file exists, file missing, invalid path.

Expected Output #3:

- Safe file handling with exception management.

PROMPT:

#Analyze this Python code that crashes when a file does not exist: `def read_file(filename): with open(filename,'r') as f: return f.read()`

#Fix the runtime error using try—except, add a user-friendly error message, and include 3 assert tests for file exists, file missing, and invalid path cases.

code:

```
def read_file(filename):
```

```
    try:
```

```
        with open(filename, "r") as file:
```

```
            return file.read()
```

```
    except FileNotFoundError:
```

```
        return "File not found."
```

```
print(read_file("nonexistent.txt"))
```

```
# Assert tests assert read_file("nonexistent.txt") == "File  
not found."
```

```
# To test the function with an existing file, we can create a temporary file and write to it
```

```
import tempfile
```

```
with tempfile.NamedTemporaryFile(delete=False) as tmp:
```

```
    tmp.write(b"Hello, World!") tmp_filename = tmp.name
```

```
assert read_file(tmp_filename) == "Hello, World!"
```

```
import os os.remove(tmp_filename)
```

Test with an empty file with

tempfile.NamedTemporaryFile(delete=False) as tmp:

tmp_filename = tmp.name

assert read_file(tmp_filename) " os.remove(tmp_filename)

output :

```
A7.1.py > ...
67 def read_file(filename):
68     return "File not found."
69 print(read_file("nonexistent.txt"))
70 # Assert tests
71 assert read_file("nonexistent.txt") == "File not found."
72 # To test the function with an existing file, we can create a temporary file and write to it
73 import tempfile
74 with tempfile.NamedTemporaryFile(delete=False) as tmp:
75     tmp.write(b"Hello, World!")
76     tmp_filename = tmp.name
77 assert read_file(tmp_filename) == "Hello, World!"
78 import os
79 os.remove(tmp_filename)
80 # Test with an empty file
81 with tempfile.NamedTemporaryFile(delete=False) as tmp:
82     tmp_filename = tmp.name
83     assert read_file(tmp_filename) == ""
84     os.remove(tmp_filename)
85
```

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```
/usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/LAB Assignments/A7.1.py"
saivenkatesh@Sais-MacBook-Air LAB Assignments % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/LAB Assignments/A7.1.py"
File not found.
saivenkatesh@Sais-MacBook-Air LAB Assignments %
```

Explanation:

The program crashes because it tries to open a file that does not exist, which raises a `FileNotFoundError`. By using a `try—except` block, the exception is caught and handled safely instead of stopping the program. This allows the function to return a userfriendly error message while continuing execution.

Task Description #4 (Calling a Non-Existent Method)

Task: Give a class where a non-existent method is called (e.g., `obj.undefined_method()`).

Use AI to debug and fix.

Bug: Calling an undefined method

```
class Car:
    def start(self):
```

```
        return "Car started"
```

```
my_car = Car()
```

```
print(my_car.drive())      #
```

drive() is not defined

Requirements:

- Students must analyze whether to define the missing method or correct the method call.
- Use 3 assert tests to confirm the corrected class works.

Expected Output #4:

- Corrected class with clear AI explanation.

Prompt used:

#Detect and fix the error in this Python code where a nonexistent method is called: class Car: def start(self): return "Car started" my_car = Car(); print(my_car.drive()) #Explain the cause of the error, decide whether to define the missing method or correct the method call, provide corrected code, and include 3 assert tests to verify the class works properly.

Code :

class Car:

```
    def start(self): return
```

```
        "Car started"
```

```
    def drive(self):
```

```
        return "Car is driving"
```

```
my_car = Car() print(my_car.drive())
```

```
# Assert tests assert my_car.start() == "Car
```

```
started" assert my_car.drive() ==
```

```
"Car is driving"
```

```
another_car = Car() assert another_car.start()
```

```
"Car started" assert another_car.drive() "Car
```

```
is driving" output:
```

explanation:

The error occurs because the method drive() is called on the Car object, but it is not defined in the Car class, which raises an

Task Description #5 (TypeError — Mixing Strings and Integers in Addition)

Task: Provide code that adds an integer and string 5" + 2) causing a TypeError.

Use AI to resolve the bug.

Bug: TypeError due to mixing string and integer def

```
add_five(value): return value + 5 print(add_five("10"))
```

Requirements:

- Ask AI for two solutions: type casting and string concatenation.
- Validate with 3 assert test cases.

Expected Output #5:

- Corrected code that runs successfully for multiple inputs.

Prompt used:

```
#Detect and fix the TypeError in this Python code caused by adding a string and an integer:
def add_five(value): return value + 5 ; print(add_five("10"))#Explain why the error occurs,
provide two solutions (one using type casting for numeric addition and one using string
concatenation), and include 3 assert tests to verify both solutions work correctly.
```

Code:

```
# Detect and fix the TypeError in this Python code caused by adding a string and an integer:
```

```
# def add_five(value): return value + 5 ; print(add_five(" 10"))
```

```
# The error occurs because the code is trying to add an integer (5) #
```

```
to a string (" 10"), which is not allowed in Python.
```

```
# Solution 1 : Using type casting for numeric addition
```

```
def add_five_numeric(value):
```

```
    return int(value) + 5
```

```
print(add_five_numeric(" 10" ))
```

```
# Assert tests for numeric addition assert add_five_numeric("
```

```
I O") 15
```

```
    add_five_numeric("0") == 5
```

```
    add_five_numeric("-5") == -5
```

```
assert 5 assert 0
```

```
# Solution 2: Using string concatenation def
```

```
add_five_string(value):
```

```
    return value + "5"
```

```
print(add_five_string("10"))
```

```
# Assert tests for string concatenation assert
```

```
    _ add_five_string(" 10") " 105"
```

```
assert add_five_string("Hello") "He1105"
```

```
assert _ add_five_string("") _
```

```
"5"
```

```
output:
```

```
Explanation:
```

The error occurs because Python cannot add a string and an integer, which raises a `TypeError`. One solution is to convert the input to an integer

using type casting so numeric addition can be performed. Another solution is to convert the value to a string and use string concatenation instead, depending on the intended behavior.

```
115 #def return value + 5 ; print
```

```
116 #Explain why the error occurs, provide two solutions (one using type casting for numeric additi0[
```

```
117 # The error occurs because the code is trying to add an integer (5) to a string which is 118 #
```

```
Solution 1: Using type casting for numeric addition
```

```
119     def      :
```

```
120         return int(value) + 5
```

```

121     print ( add_f ive_numeric 10" ) )
122     # Assert tests for numeric addition
123     assert add_five_numeric( " 10 " ) --- 15
124     assert add_five_numeric(      )
125     assert add_five_numeric("-5")
126     # Solution 2: Using string concatenation 127     def add_five_string(vatue) :
128     return value + "5"
129     print ( add_f ive_string ( )
130     # Assert tests for string concatenation
131     assert "105
132     assert add_f ive_st ring ("Het )      "He1105"
133     assert add_five_string( 134

```

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/usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/LAB Assignments/A7.1.py"

- saivenkatesh@Sais—MacBook—Air LAB Assignments % /usr/bin/python3 "/Users/saivenkatesh/Desktop/AI Coding/L 15
105